

To: Leadership and Product Management, STADIOEquities

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Subject: Recommendations on Predictive Modelling for Client Activation Interventions

In this report we present the results of our recent experiments related to predictive modeling. In particular, we are going to present and discuss our findings regarding machine learning algorithms intended to predict which users are going to need the intervention in order to undergo the desired financial conversion. In the light of the evaluation of our experimental models, this report suggests strategic recommendations concerning the choice of the most suitable algorithm, its optimization approaches, adaptation to the STADIOEquities activation funnel, and the conformity with the existing domain literature.

1. Model Selection: Random Forest vs. XGBoost

Based on the rigorous statistical comparison of Model 1 (eXtreme Gradient Boosting) and Model 2 (Random Forest), we can clearly conclude that the latter is definitely better architecture for our specific business purpose of predicting minority-class conversions.

Even though XGBoost outperformed Random Forest in terms of global accuracy (88.52% against 86.74%), accuracy in such strongly unbalanced datasets is a statistically misleading parameter. In our business case we aim at minimization of the number of false negatives that means missing the identification of a prospective client who needs an onboarding intervention to be converted.

The Random Forest model with the balance parameter set to 1 has succeeded in moving the prediction threshold of the model in favor of the minority class, thus achieving a better Recall value of 0.62, which means that the model has been able to catch 62% of actual cases against 51% in XGBoost (Recall: 0.51). Moreover, the model of Random Forest achieved a higher Area Under the Receiver Operating Characteristic Curve (ROC-AUC) score of 0.8077 against 0.7994 in XGBoost. It proves that across different operation thresholds the Random Forest model has the greater ability to discriminate between the clients who are going to convert naturally and the ones who are going to stay stalled.

2. Options for Improving the Model

Despite the evident viability of the Random Forest baseline, implementation of the model in live production requires some improvements in order to improve the ROI of acquisition. In particular, the following changes are recommended:

Hyperparameters Tuning: The existing model relies on a fixed baseline of hyperparameters. The usage of GridSearchCV will help to test different options for depth, minimum number of samples in a leaf, and maximum features considered at each step in order to gain some extra predictive power without causing overfitting.

Probability Threshold Tuning: Most classification models use the 0.5 probability as a cutoff value. In our case, the cost of a false positive (sending an email with a standard offer) is much

lower than the cost of a false negative (missing the opportunity to invest R180 in acquisition). Thus, precision-recall curve analysis can help us to tune down the cutoff probability to 0.35 or 0.40, causing the purposeful increase in the number of true positives.

Improved Smote-Based Resampling: Though SMOTE proved to be quite effective in balancing the training set, it can sometimes generate noisy synthetic data. In order to mitigate the precision reduction (0.44) in the existing model, the combination of SMOTE and ENN (SMOTE-ENN) will be used.

In order to effectively apply the proven Random Forest approach to tackle the problem of STADIOEquities “activation gap” (funded deposit prediction), the data flow needs to be adjusted, yet the underlying logic of the algorithm itself should remain untouched.

The first thing to do in this case is to switch the feature space from traditional demographic attributes to modern digital behavior metrics. In this case, the model will need to consume granular data on app sessions, KYC process timestamps, funding delays, and particular acquisition channel features.

The second thing to adjust about the data engineering flow is to tackle the problem of unified identity found in SS1 risk log. Since in case of STADIOEquities customers’ data are stored in different sources – web, mobile apps, marketing systems – it’s necessary to build a SQL layer in order to build a unified client_id before passing it into the Random Forest model.

Lastly, the model deployment architecture should shift from batch processing to near-real-time API predictions in order to reduce the 31% customer dormancy rate at an early stage.

The empirical results observed in our experiments align seamlessly with established academic literature concerning predictive modeling in retail banking, though it is necessary to acknowledge the fundamental differences between our SS2 experimental data and the final Part A environment.

Our SS2 models utilized the UCI Bank Marketing dataset as a proxy, relying on traditional telemarketing features (e.g., call duration, previous campaign outcomes) to predict *term deposit subscriptions*. Conversely, the STADIOEquities environment (Part A) requires predicting *first-deposit activation* using modern, digital app behavioral data.

Despite these distinct data domains, the underlying mathematical challenge—identifying rare, high-value financial conversions amid severe class imbalance—remains identical. Consequently, our findings strongly corroborate Moro, Cortez, and Rita (2014), who demonstrated that tree-based ensemble methods successfully capture non-linear relationships in financial conversion datasets better than traditional logistic regression. Furthermore, our reliance on ROC-AUC and Recall over raw accuracy validates the consensus in imbalanced learning literature (Chawla et al., 2002), confirming that our Random Forest methodology is theoretically sound and highly transferable to the STADIOEquities application.

References

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